Master’s Thesis Summary

Comparative Analysis of Image-Based AI Methods for Fault Classification in Thermal Adhesive Application for High-Voltage Battery Modules

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1. Executive Summary

This master's thesis investigates how image-based artificial intelligence can support the automated classification and analysis of defects in the application of thermally conductive adhesive used in high-voltage battery modules. The work is motivated by a common industrial challenge: automated inspection systems can flag an application as defective, but the subsequent categorization and interpretation of the fault often still require substantial manual effort.

The study develops and compares several analytical approaches rather than focusing on a single model. These include a supervised deep-learning approach based on YOLO, an unsupervised anomaly-detection pipeline using learned image representations and nearest-neighbour reasoning, a geometric rule-based approach, and a hybrid approach combining image information with selected process-related features. The approaches are assessed under a shared evaluation framework to examine classification quality, robustness, processing speed, computational efficiency, interpretability, and suitability for deployment in a production-oriented environment.

The central contribution is a structured comparison of fundamentally different solution strategies. The thesis does not treat model accuracy as the only success criterion. Instead, it examines the trade-offs between predictive performance, stability across defect categories, explainability, implementation effort, and operational practicality. This creates a decision basis for selecting an appropriate inspection and analysis method according to the requirements of a specific manufacturing use case.

2. Industrial and Technical Context

Thermally conductive adhesive is used to create a reliable thermal and mechanical interface between battery cells and cooling components. The applied adhesive pattern must satisfy geometric and quality-related requirements so that heat can be transferred effectively and the assembly remains mechanically stable. Variations in application quality may arise from material behaviour, dispensing conditions, component positioning, contamination, equipment settings, or image-acquisition effects.

Modern manufacturing lines typically use camera-based inspection to monitor the applied adhesive. Such systems are effective at detecting deviations from expected patterns, but a binary good/not-good decision provides limited support for deeper process learning. Engineers also need to understand what type of defect occurred, whether similar faults recur, and which analytical method can classify the fault consistently and quickly.

The thesis therefore addresses the gap between defect detection and defect understanding. Its focus is not on replacing an existing safety-relevant inspection decision. Instead, the work explores a secondary analysis layer that can structure defect images, classify recurring patterns, compare alternative AI methods, and support more systematic technical evaluation.

3. Research Aim and Questions

The overall aim is to determine which image-based analytical approach is most suitable for automated fault classification and process-oriented analysis in a thermal adhesive application use case.

- How effectively can different image-based AI methods classify recurring adhesive-application fault patterns?

- How do the supervised image-only YOLO approach, the supervised hybrid YOLO and process-parameter approach, the unsupervised anomaly-detection approach, and the rule-based approach differ in precision, recall, F1-score, robustness, and runtime?
- Which optimization measures improve each method without undermining comparability or causing excessive methodological drift?
- What trade-offs arise between predictive quality, transparency, data requirements, computational effort, and deployment suitability?
- How can the results be translated into practical guidance for production analysis and future process improvement?

4. Methodological Framework

The research follows an applied comparative design. A common evaluation dataset and harmonized performance indicators are used wherever possible so that differences between the methods can be interpreted meaningfully. The workflow comprises data understanding, preparation, model development, optimization, testing, KPI calculation, robustness analysis, and cross-method discussion.

4.1 Data Preparation and Ground Truth

The starting point is a collection of image records from an automated adhesive-inspection process. The data-preparation stage includes quality checks, removal of unusable records, consistent file handling, review of visual patterns, and creation of a reference classification for evaluation. Because industrial defect datasets are often imbalanced, the thesis also considers the effect of class distribution and data balancing on model behaviour.

Production and defect images are not included in this summary. Other dataset and process details may be discussed, subject to applicable disclosure approval; details not supplied here have not been added.

4.2 Common Evaluation Logic

Each approach is evaluated with metrics derived from true positives, false positives, and false negatives. Precision indicates how reliably a predicted defect class is correct. Recall indicates how many relevant cases are retrieved. The F1-score balances precision and recall and is especially useful where class frequencies differ. Processing time and images processed per second capture operational efficiency. Robustness is assessed through performance consistency across classes or scenarios, with attention to weak classes rather than only average behaviour.

5. Approaches Developed and Compared

Approach	Core idea	Main strength	Main limitation
Supervised YOLO-based model	Learns defect categories from labelled images.	Direct multi-class prediction and strong task-specific learning.	Requires representative labels and can be sensitive to class imbalance.
Anomaly detection	Compares learned image features with a reference memory and groups deviations using nearest-neighbour and clustering methods.	Can discover structure with less dependence on exhaustive class labels.	Thresholds and clusters require careful interpretation and validation.

Rule-based geometric analysis	Uses measurable shape or geometry criteria to identify deviations.	Transparent decisions and easy engineering interpretation.	Limited flexibility when defects are visually complex or ambiguous.
Supervised hybrid YOLO and process-parameter approach	Combines YOLO-derived visual information with selected process parameters in a labelled supervised-learning framework.	Uses both image evidence and process context for fault classification.	Requires synchronized, representative image and process data and has higher integration complexity.

5.1 Supervised Image Classification

The supervised branch uses a YOLO-family model to learn distinctive visual characteristics of predefined fault categories. Work in this branch includes dataset structuring, training and validation, class balancing, confidence-related settings, and evaluation of errors at class level. The analysis emphasizes whether improvements are broad and repeatable rather than driven by one dominant category.

5.2 Anomaly Detection and Representation Learning

The anomaly-detection branch extracts numerical feature representations from images and compares them with a reference memory bank. Distances to the memory provide an anomaly signal, while nearest-neighbour assignment and clustering support the grouping of recurring patterns. The optimization work examines data cleaning, decision thresholds, dimensionality reduction with UMAP, density-based clustering with HDBSCAN, and the balance between local separation and global stability.

This branch is particularly valuable when defect labels are incomplete, evolving, or expensive to create. Its main methodological challenge is that a mathematically coherent cluster does not automatically correspond to an engineering-relevant defect category. Human validation therefore remains important.

5.3 Rule-Based Analysis and Supervised Hybrid Classification

The rule-based approach translates engineering expectations into explicit geometric checks. This provides traceable decisions and a useful baseline. The hybrid approach extends the visual assessment with selected process-related context. Both approaches broaden the comparison, but the final optimization emphasis is placed on the strongest image-based contenders so that the study remains focused and the optimized scenarios remain directly comparable.

6. Optimization Strategy

Optimization is conducted as a controlled series of scenarios. Instead of changing many pipeline components simultaneously, each scenario isolates a defensible adjustment, such as data balancing, feature-space configuration, clustering choice, neighbourhood settings, or class-decision thresholds. This supports attribution: when performance changes, the likely cause is easier to identify.

A key principle is methodological stability. The original pipelines are preserved wherever possible, and optimization is treated as refinement rather than wholesale redesign. This protects comparability between baseline and optimized configurations and reduces the risk that a better score is achieved through an entirely different method.

7. Results at a Non-Confidential Level

The comparative work shows that no single method is universally superior across all criteria. Supervised learning is well suited to direct classification when sufficient representative labels are available. Anomaly detection offers flexibility for emerging or weakly labelled patterns and can reveal useful structure in the feature space. Rule-based analysis remains valuable where geometry is well defined and explainability is critical. Hybrid approaches can add context but require more complex data integration and governance.

The optimization phase demonstrates that model selection should not rely on the highest average score alone. A configuration with excellent precision may overlook relevant cases, while a high-recall configuration may produce too many false alarms. Similarly, a pipeline that performs well overall may still fail on a minority class. The thesis therefore gives special attention to class-level performance, weakest-class behaviour, and the relationship between quality and processing time.

The final method comparison concentrates on the optimized supervised and anomaly-detection configurations because these emerge as the most promising candidates for deeper evaluation. The rule-based and hybrid methods retain value as reference approaches and as possible components of future combined solutions.

8. Scientific and Practical Contribution

- A common evaluation framework for comparing heterogeneous inspection approaches under consistent criteria.
- A structured view of trade-offs between supervised image-only classification, supervised hybrid classification using YOLO and process parameters, unsupervised anomaly detection, and explicit engineering rules.
- A scenario-based optimization method that preserves traceability and scientific comparability.
- An emphasis on robustness and weakest-class performance, not only average model scores.
- Practical guidance for matching a method to data availability, explainability needs, runtime constraints, and expected defect evolution.
- A foundation for more systematic defect analytics, recurring-pattern analysis, and targeted process investigation.

9. Limitations

The study is based on one industrial application domain, so conclusions should be transferred carefully to other processes. Performance depends on the representativeness and quality of the available images and labels. Rare faults remain difficult to evaluate statistically, and changes in lighting, viewpoint, material appearance, or equipment can affect model behaviour. Unsupervised clusters require engineering interpretation, while supervised methods require maintenance when fault definitions evolve. The thesis therefore treats the models as decision-support tools that require validation, monitoring, and clear responsibility in any future operational use.

10. Future Work

- Conduct ablation studies to quantify the contribution of individual preprocessing, feature, threshold, and clustering components.
- Test the pipelines on additional datasets, time periods, or operating conditions to assess generalization and drift.
- Improve model monitoring with class-level dashboards, drift indicators, and traceable versioning of data and model configurations.

11. Plain-Language Takeaway

In simple terms, the thesis asks: once a camera system identifies that an adhesive application is not acceptable, how can artificial intelligence help explain what kind of problem occurred and organize similar cases efficiently? To answer this, the work builds and compares different ways of analysing defect images. Some methods learn from labelled examples, some identify unusual patterns by comparing images with known references, and others use explicit engineering rules or additional contextual data. The outcome is not merely a “best model,” but a reasoned framework for choosing the right method for the task, available data, and operational constraints.

12. Short Glossary

Term	Meaning
Precision	Share of predicted cases that are correct.
Recall	Share of relevant cases that are successfully found.
F1-score	Harmonic balance of precision and recall.
Anomaly detection	Identification of samples that differ from a learned or defined reference.
Memory bank	Stored reference feature representations used to compare new images.
UMAP	A dimensionality-reduction method used to represent complex feature spaces more compactly.
HDBSCAN	A density-based clustering method that can identify clusters of different shapes and mark uncertain points as noise.
Robustness	Consistency of performance across classes, conditions, or evaluation scenarios.